

A) Selecting from the database

Exercise 1)

- i. Create a new class modelling the products table (Product.java) – this is a DTO (Data Transfer Object). Like the Customer class in the example, this class should contain variables based on the table as well as constructors, getters (no setters) and the "good practice" methods.
- ii. Create a program that:
 - a. Opens a connection to the classicmodels database
 - b. Reads in all entries in the products table and stores them as Product objects in an arraylist.
 - c. Displays the contents of your arraylist.

Exercise 2)

- i. Amend the Product class to implement the Comparable interface. Products should be ranked in order of product code (DESCENDING).
- ii. Create a program that:
 - Requests a product line from the user and selects all Products containing that product line from the database. These should be stored as Product objects in an arraylist.
 - Displays the results in descending order of product code.

Exercise 3)

- i. Create a program that takes in a keyword from the user and displays the details of all products with a name containing that keyword.

B) Adding to and removing from the database

Note: **executeUpdate()** should be used for update, delete and insert SQL operations. It will return a number indicating how many rows were affected by the operation (e.g. the number of rows changed by an update or the number of rows deleted by a delete operation)

Exercise 4)

Create a program that:

- i. Takes in the information to create a new Product object and inserts the new Product into the products table.
- ii. Displays the updated list of Products from the database.

Exercise 5)

Create a program that:

- i. Requests a product vendor from the user and selects all products sold by that vendor from the database.
- ii. Displays the products in order of product code (DESCENDING) (re-ordering the information should be done using the sort() from ArrayList).
- iii. Asks the user if they wish to delete any products – if so, delete the corresponding entry from the database and display a message indicating success/failure.
- iv. Selects all products matching the originally specified product vendor (taken in during part i.) and display in order of product code (DESCENDING).

Customer Admin Application

Exercise 6)

- i. Add a users table to the database – this should include the following attributes:
 - username (primary key)
 - password
 - email address (unique field)
 - usertype (admin/user)

The table should be well-structured, i.e. appropriate type used for each field, marking fields as unique where appropriate.

- ii. Add 4 rows of dummy data to the users table
- iii. Create a User DTO (a User.java class based on the users table).
- iv. Create a program that asks the user to login or register

Remember:

- Logging in is essentially checking if there is an entry in the appropriate table with the same username AND password.
 - Registering is inserting a new entry into the database
- v. Upon successful login/registration, the application should have two modes – admin and user.
 - Admins can view, update and delete products
 - Users can view products and update their own user information.